

Spectrum

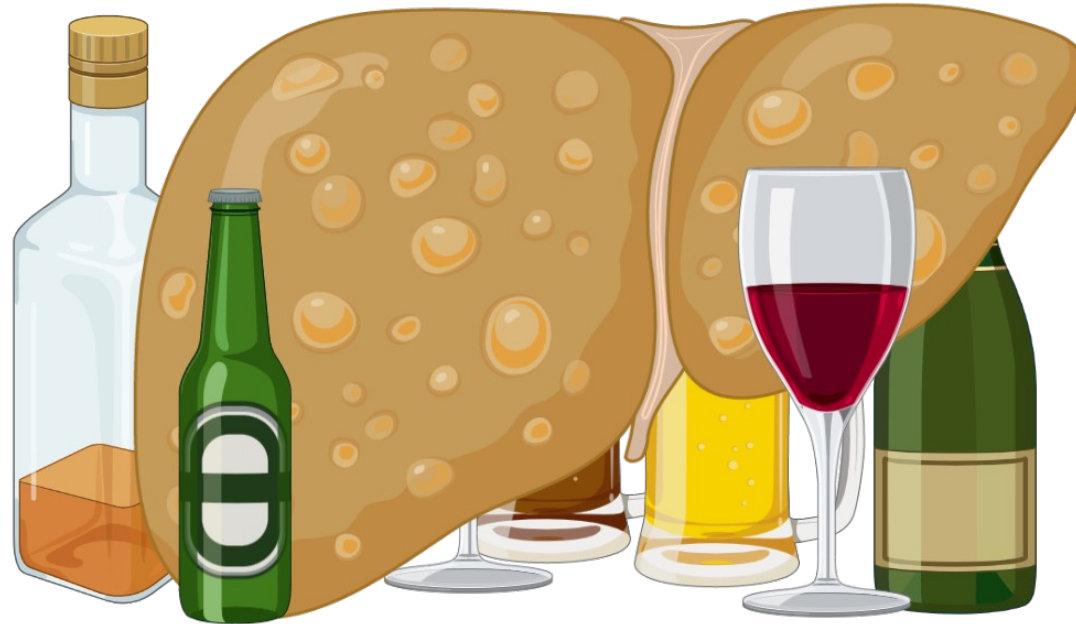
Diagnosis

Management

AH

Cases

Alcohol-related Liver Disease



Spectrum

Diagnosis

Management

AH

Cases

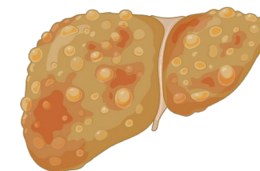
Risk factors



- Volume (M: ≥ 3 /d | F: ≥ 2 /d)
- Pattern (e.g., binging)



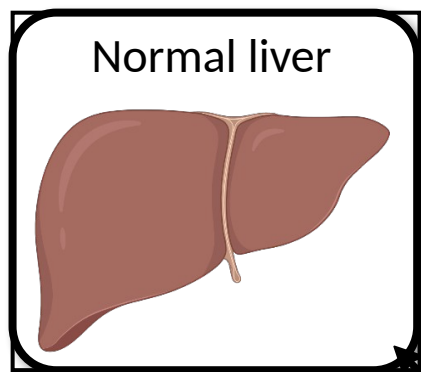
Genetics



Other liver disease

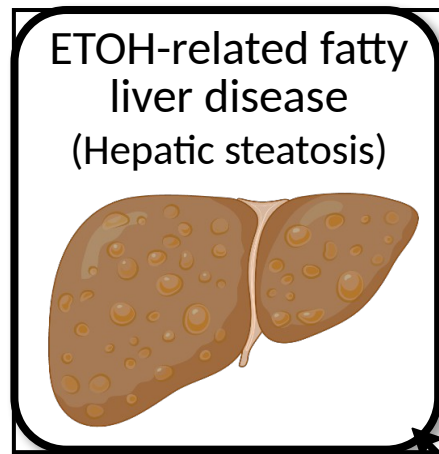
- Viral hepatitis
- NASH

ETOH use



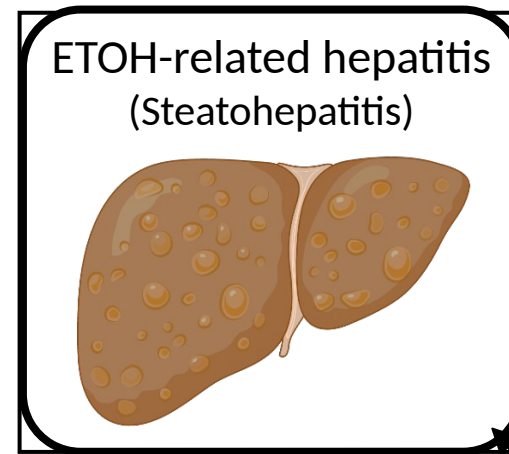
Normal liver

ETOH use



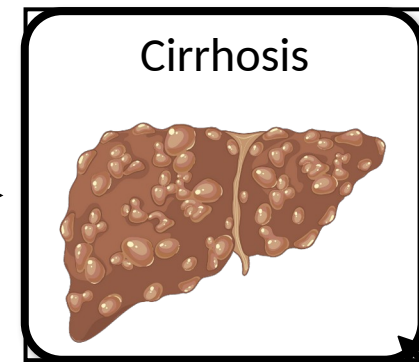
ETOH-related fatty liver disease (Hepatic steatosis)

ETOH use



ETOH-related hepatitis (Steatohepatitis)

ETOH use



Cirrhosis

Acute onset symptomatic hepatitis

Acute decompensation occurs with ongoing ETOH use

Overview / Spectrum of Disease

Overview

Diagnosis

Management

AH

Cases

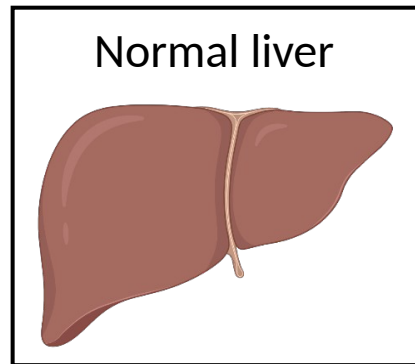
Diagnostic Evaluation

Who to screen?

Screen all > 18 yo with unhealthy
ETOH use

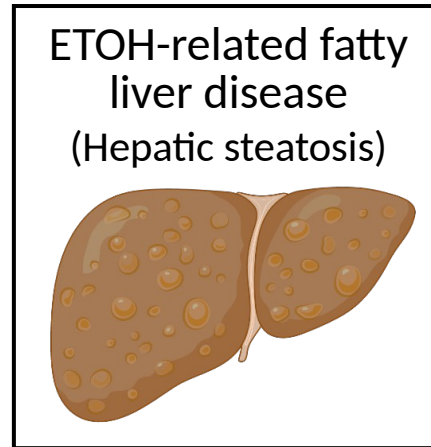
How to screen?

RUQ US
LFTs, CBC, INR



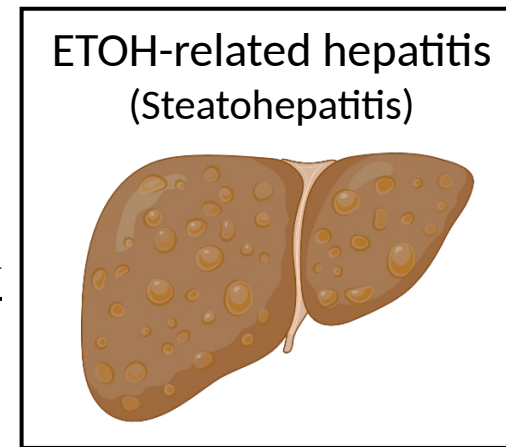
Normal liver

ETOH use



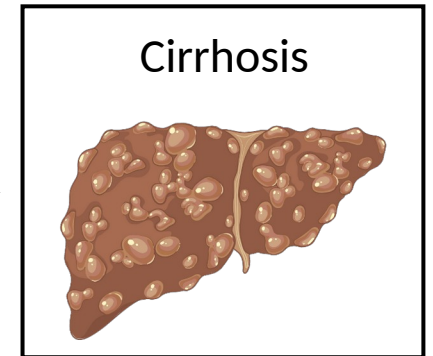
ETOH-related fatty
liver disease
(Hepatic steatosis)

ETOH use



ETOH-related hepatitis
(Steatohepatitis)

ETOH use



Cirrhosis

Diagnosis

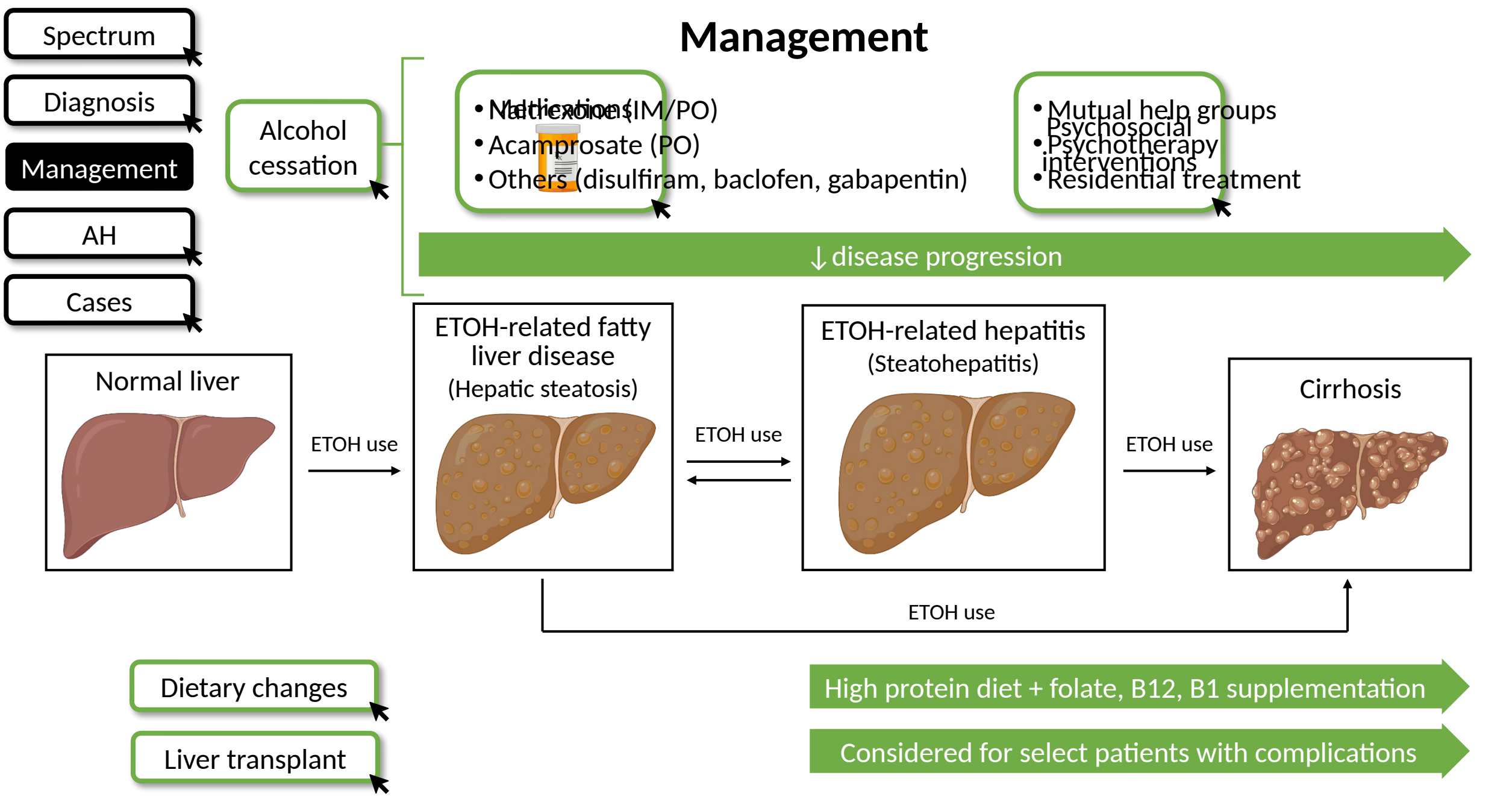
- Often incidental finding on imaging
- $\pm$ $\uparrow$ LFTs
- Indistinguishable from NAFLD on US or biopsy

Diagnosis

- Clinical diagnosis!
- Typically, AST:ALT is > 2
- $\uparrow$ LFTs increases likelihood of steatohepatitis

Diagnosis

- Risk scores (e.g, APRI, FibroTest, ELF)
- Small nodular liver on US
- Elastography
- Liver biopsy



Diagnosis of Alcohol-related Hepatitis (AH)

Spectrum

Diagnosis

Management

AH

Diagnosis

Treatment

Cases

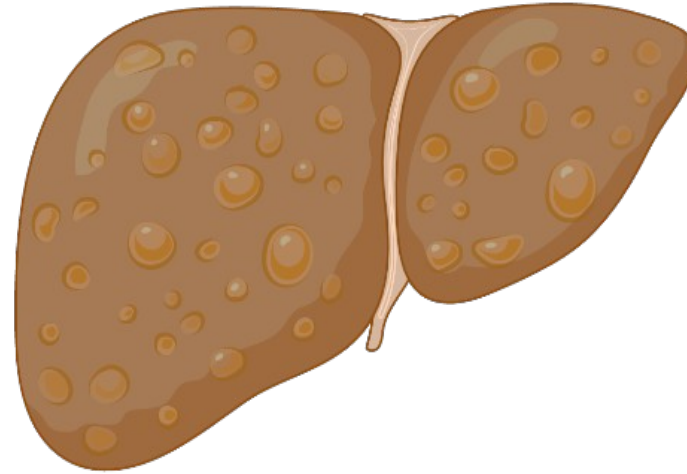
Clinical features

- Fever
- RUQ pain
- Anorexia
- Ascites
- **Jaundice (≤ 8 wks)**

- Heavy, recent ETOH use

 >4  /d |  >3  /d

<60 d of abstinence prior to onset of jaundice



Diagnostic work-up

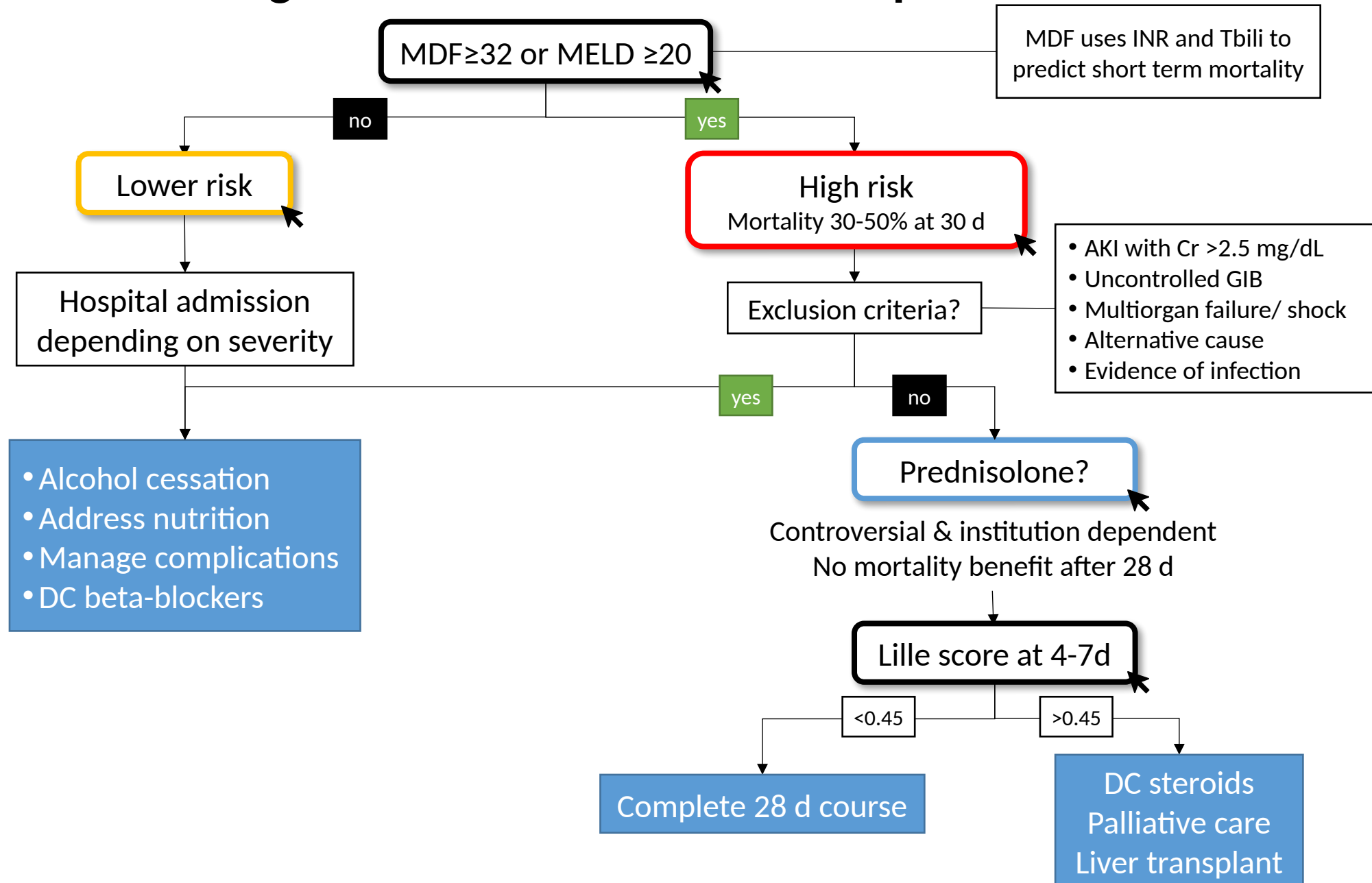
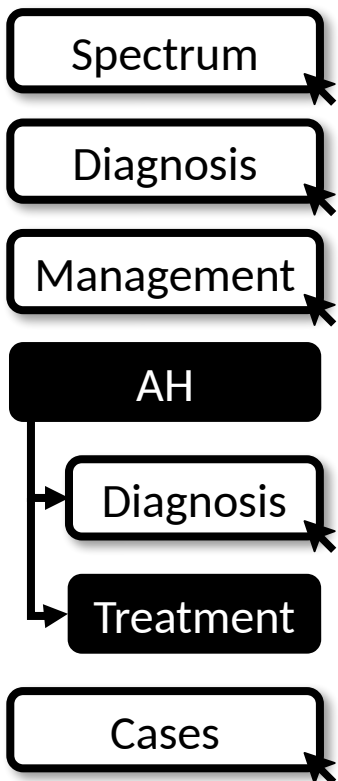
- Rule out other causes of hepatitis
 - Acute hepatitis panel
 - Drug screen and APAP level
 - RUQ US with doppler
- Liver biopsy (rarely needed)

Laboratory findings

- **$\uparrow$ AST:ALT >2** (usually <400 IU/L)
- **$\uparrow$ Total bilirubin >3 mg/dL**
- $\uparrow$ GGT
- $\uparrow$ WBC (PMN predominant)
- $\uparrow$ INR

AASLD Guideline for clinical diagnosis of AH

Management of Alcohol-related Hepatitis



Spectrum

Diagnosis

Management

ETOH hep

Cases

1

2

Case 1a

60-year-old M with history of HTN presents for an annual wellness visit. His only medication is lisinopril 10mg. He drinks a 6-pack of beers every night. He is a never smoker and has no history of drug use.

Exam: BP 129/78, P 75, SpO2 94%, BMI 28. Physical exam is notable for hepatomegaly.

Labs:

- WBC 8, Hgb 14, plts 208
- AST 80, ALT 55, ALP 110, Tbili 0.9, and INR 1.1.
- Negative HCV antibody within the past year.

What are the next best steps for evaluation/management of his abnormal LFTs?

A) RUQ US

B) Repeat labs in 3-6 months

C) Discuss alcohol cessation

D) Refer to Hepatology

Case 1b

He is abstinent for 3 months and returns to see you in clinic.

Labs:

- WBC 8, Hgb 14, plts 108
- AST 87, ALT 65, ALP 118, Tbili 0.9, and INR 1.5.

What are the next best steps for evaluation of his abnormal LFTs?

A) RUQ US

B) Repeat labs in 3-6 months

C) Calculate APRI

D) Refer to Hepatology

...RUQ US shows a small, nodular liver with splenomegaly. No ascites is seen.

What is your next step in management?

- Refer to Hepatology.
- Nutritional counseling (high protein diet, vitamin supplementation)
- Continued abstinence counseling.
- Health maintenance (viral hepatitis testing and vaccination, EGD for varices, routine HCC screening)

Overview

Diagnosis

Management

ETOH hep

Cases

1

2

Case 2a

35-year-old F presents with 1 week of progressive jaundice, new onset ascites, and dyspnea on exertion. She drinks on average 3-4 alcohol containing drinks per night and occasionally binge drinks. Last drink ~24 h ago. No other substance use. No medications or supplements.

Exam: T 37.9C, HR 90, BP 120/88, RR 20, SpO2 94% on RA. Jaundiced, has a distended abdomen with fluid wave present, and 2+ lower extremity edema extending to the knees bilaterally. Confused with asterixis.

Labs:

- Cr 0.82, BUN 11, WBC 14, Hgb 11, plts 108
- AST 224, ALT 110, ALP 118, Tbili 3.4, and INR 2.1.
- RUQ US with enlarged fatty liver and moderate volume ascites.

What is the most likely diagnosis?

A) Decompensated cirrhosis

B) Alcohol-related steatosis

C) Spontaneous bacterial peritonitis

D) Alcohol-related steatohepatitis

Case 2b

Evaluation for infection including CXR, UA, and SBP is negative.

What is the next best step in management?

The **MDF is 94**, which is associated with high short-term mortality. Given that there are no contraindications (normal renal function, no GIB, infection ruled out), then **steroids could be considered**.

...She is started on prednisolone 40 mg daily.

When should she be re-evaluated for response to therapy?

Repeat CMP and coags should be obtained at **4-7 days** after starting steroid therapy to assess response. The **Lille score** can help determine response to and duration of therapy.

...After 7 days, repeat labs show:

- Cr 0.78, BUN 8, WBC 12, Hgb 11, plts 98
- AST 204, ALT 78, ALP 130, Tbili 4.4, and INR 1.8 (PT 20).
- Albumin 2.7

What is the next best step in management?

Lille score is <0.45, this suggests good prognosis and response to steroids. Six-month survival is 85%. **Steroids should be continued for 28 d total.**

Guideline & Nomenclature Update (2024)

2024 ACG Clinical Guideline on Alcohol-Associated Liver Disease (Am J Gastroenterol 2024;119:30-54)

- Reaffirms the framework taught here: abstinence is the single most effective intervention; MDF ≥ 32 (or MELD >20) identifies severe alcohol-associated hepatitis (AH) for corticosteroid consideration; Lille score at day 4–7 guides whether to continue or stop.
- Prednisolone 40 mg/day improves 28-day (short-term) survival in severe AH without contraindications — no long-term benefit. Adding IV N-acetylcysteine (NAC) to prednisolone may further improve short-term (28–30 day) survival.
- Early liver transplantation for carefully selected patients with severe AH non-responsive to medical therapy is now endorsed by ACG/AASLD (no longer experimental), with outcomes comparable to other indications.
- 2023 multi-society nomenclature change (Hepatology 2023;78:1966): NAFLD $\rightarrow$ MASLD, NASH $\rightarrow$ MASH; new "MetALD" category for metabolic dysfunction plus moderate alcohol use. "Alcohol-related steatosis is indistinguishable from MASLD on imaging" replaces the old NAFLD wording on the diagnosis slides.

Standing teaching slides are unchanged and remain current; this slide captures the newer framing and adjuncts.

Take-Home Points

- ALD is a spectrum: hepatic steatosis → alcohol-related (steato)hepatitis → fibrosis/cirrhosis. Steatosis and inflammation can regress with abstinence; fibrosis is largely irreversible.
- Abstinence is the single most important factor improving survival. Use a multimodal approach: naltrexone or acamprosate first-line (FDA-approved), plus psychosocial/behavioral treatment. Avoid disulfiram in ALD.
- Alcohol-related hepatitis (AH) is a CLINICAL diagnosis: heavy recent use, jaundice ≤ 8 wks, AST:ALT > 2 with both < 400 IU/L, total bilirubin > 3 mg/dL. Rule out infection and other causes in every patient.
- Risk-stratify AH with MDF (≥ 32) or MELD (> 20). Consider prednisolone 40 mg/day ($\pm$ NAC) if no contraindication (active infection, AKI Cr > 2.5 , GI bleed, multiorgan failure). Reassess with Lille at day 4–7: < 0.45 = continue 28-day course; ≥ 0.45 = stop steroids.
- Nutrition matters: high-protein diet with thiamine, folate and B-complex supplementation. Refer non-recovering cirrhosis — and selected severe AH — for transplant evaluation.

Attribution, License & Changelog

Adapted from the TeachIM original "Alcohol-related Liver Disease" (published November 2022).

- Authors: CJ Mowry, MD; John Haydek, MD. Section Editor: Elijah Mun, MD. Executive Editor: Yilin Zhang, MD.
- Source: https://teachim.org/teaching_material/alcohol-related-liver-disease/
- License: Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0). Non-commercial educational use only.
- Optimization (June 2026): surgical correction — Case 2b slide "Six-month mortality is 85%" → "survival is 85%" (Lille <0.45 predicts ~85% 6-month survival; Louvet, Hepatology 2007). All original click-to-reveal animations and BioRender graphics preserved.
- Added: 2024 ACG ALD guideline update + 2023 MASLD/MASH nomenclature slide; take-home slide. Companion files: ALD facilitator guide (ALD-teaching-script.pdf), ALD learner handout (ALD-handout.pdf) — reviewed, no changes needed.

Medical content should be clinician-reviewed before teaching.